TRI-OCTAGON MATHEMATICAL RESEARCH

A Reciprocal Chiral Exchange Extension of the Tri-Octagon Model

Well-posedness, matched storage, and extraction limits

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Abstract

We study one explicit candidate extension coupling the Tri-Octagon native state to its conditional periodic carrier and to a reversible two-port receiver. The existing raw chiral drive is retained, while a homogeneous reciprocal completion matches carrier work to changes in a positive trial native storage. A quadratic receiver storage has an equal and opposite carrier loading term. We prove global forward and backward exchange-time well-posedness in the natural energy space, with a separate regularity estimate for periodic H^1 preparations. Empty-carrier preparations exhibit nonzero short-time native supply and receiver gain; other prepared carrier states yield native credit. The exchange conserves the complex bilinear quantity $\sum_i \Omega_i^2$, imposing a restrictive bound on net extraction from coherent initial states and leaving aligned empty preparations dark. Explicit counterexamples prevent transfer of universal coherent-domain invariance to the coupled model. An ordered hybrid step retains the unchanged native map before exact exchange; its total storage increment is precisely the native baseline intensity increment, and zero coupling recovers the original native sampling. These are results about a mathematically verified candidate, not adoption into the accepted kernel or a derived physical energy law. Delayed native credit on the same solution after an empty launch, irreversible disposal and physical toroidal registration remain open.

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1 Question, assumptions and exact candidate

The previous carrier account proved source-off recurrence and distinguished section throughput from native return and diversion. [3] It also isolated missing reciprocal debit and receiving-load equations. The present paper supplies and analyzes one proposed completion of those interfaces, following the accepted analytical investigation. [1] The proposal is not selected by the old model: its new storage, feedback, receiver and timing assumptions are part of the statement being tested. Table 1 separates these roles.

Status	Ingredients and interpretation
Inherited	Native map F, raw chiral drive and registration; conditional C1 circle, finite tents, propagation and carrier storage.
Proposed	Scaled native intensity, receiver quadratic storage, reciprocal feedback, collocated loading, continuous drive protocol and F-then-exchange cadence.
Verified here	Coupled well-posedness, matched accounts, short-time activity, prepared credit, invariant capacity and domain limitations, exact hybrid balance.
Not established	Physical energy calibration, kernel adoption, delayed return after an empty launch on the same solution, irreversible disposal, central route or six-opening allocation.

Table 1: Definitions, design choices and deductions retain different status.

Power-preserving interconnection and matching input/output power are established methodology; see van der Schaft [6, Sections 1.1–1.2, especially (5)–(6)]. That reference does not select this candidate or prove its PDE theorem. We prove the identities and existence argument directly and make no canonical Hamiltonian/Poisson or novelty claim for the general methodology.

1.1 Inherited raw chiral drive

Let $x = (a, b) \in \mathbb{R}^6$, $\Omega = a + ib \in \mathbb{C}^3$ and $I = |x|^2$. Here b is the imaginary native vector; b_σ below is a source profile and d_{pre} is the native prestage increment. Retain the source's frame, normals and fixed registration [2, foundations]:

$$u = (0, 1, 0), \quad e_z = (0, 0, 1), \quad n_\pm = \frac{2u \pm \sqrt{3}e_z}{\sqrt{7}}, \quad A_{\mathrm{reg}} = \lambda_g R_z(\delta_g). \tag{1.1}$$

The gains $\lambda_g, \kappa_{\mathrm{in}}$ are positive and fixed, and the yaw δ_g is fixed. Define

$$K = \kappa_{\mathrm{in}} \begin{pmatrix} n_+^T \\ n_-^T \end{pmatrix} A_{\mathrm{reg}}, \qquad f(x) = K(a \times b), \qquad J(x) = D_x f(x). \tag{1.2}$$

K is a constant rank-two 2×3 matrix and J is 2×6. This retains raw channel chirality, not its distinct ambient axial attachment. The inherited contact tangents are $-n_+$ and $+n_-$; those orientation signs do not insert signs into the source projection (1.2). The macro readout, its external clock and the blend are not used.

1.2 Finite carrier ports and trial storages

Retain C1 on $s \in \mathbb{R}/L_c\mathbb{Z}$, with $c > 0$. The contact positions are $s_\pm = L_c/2 \mp \eta_{\mathrm{geom}}$; their shortest separation is $2\eta_{\mathrm{geom}}$. For $0 < \varepsilon_c < \eta_{\mathrm{geom}}$, let

$$b_\sigma(s) = \varepsilon_c^{-1} \left(1 - \frac{|\operatorname{wrap}_{L_c}(s - s_\sigma)|}{\varepsilon_c}\right)_+, \qquad \sigma \in \{+, -\}. \tag{1.3}$$

These normalized disjoint tents belong to periodic H^1. Their width is arclength width, not a tube radius. Define the real operator and its adjoint

$$\mathcal{B}v = b_+v_+ + b_-v_-, \qquad y = \mathcal{B}^*p = \begin{pmatrix} \int b_+p\,\mathrm{d}s \\ \int b_-p\,\mathrm{d}s \end{pmatrix}. \tag{1.4}$$

All unqualified spatial integrals cover one whole circle. With one receiver state $z = (z_+, z_-) \in \mathbb{R}^2$, choose

$$H_N = \frac{\alpha_N}{2}|x|^2, \quad H_C = \frac{1}{2}\int (p^2 + r_c^2)\,\mathrm{d}s, \quad H_R = \frac{\beta_R}{2}|z|^2, \quad H_{\mathrm{tot}} = H_N + H_C + H_R, \tag{1.5}$$

where $\alpha_N, \beta_R > 0$. Only H_C is inherited. H_N and H_R are new constitutive choices; H_N is not the native quartic potential V. The receiver is internal to this enlarged account without an assigned spatial or material location. Its two coordinates are not two new holes or physical objects.

1.3 Exchange equations and dimensions

For dimensionless new multipliers $\eta_N, \eta_R \geq 0$, the proposed stage is

$$\boxed{\begin{aligned} \dot{x} &= -\frac{\eta_N}{2\alpha_N} J(x)^T y, \\ p_t + c(r_c)_s &= \mathcal{B}\{\eta_N f(x) - \eta_R z\}, \\ (r_c)_t + cp_s &= 0, \\ \dot{z} &= \frac{\eta_R}{\beta_R} y, \qquad y = \mathcal{B}^*p. \end{aligned}} \tag{1.6}$$

Preparation $(x_0, p_0, r_{c,0}, z_0)$ is independent; the carrier is periodic. Both couplings use the same distributed port regions. The changing native state replaces an external sample-and-hold history during this stage; the old history protocol remains a separate comparison. No coefficient is chosen as a diverted fraction.

Let A denote a bookkeeping native amplitude unit, with $[x] = \mathrm{A}$. The inherited normalized native convention sets it to one. The carrier's abstract units are U, L, T, and Q is the registered readout data unit. [2, units] Thus $[J^T y/\alpha_N] = \mathrm{A/T}$, $[y/\beta_R] = \mathrm{UL/T^2} = [\dot{z}]$, and $[\mathcal{B}f] = \mathrm{U/T}$.

Quantity	Dimension
$s, L_c, \varepsilon_c, \eta_{\mathrm{geom}};\ c;\ t, h_{\mathrm{ex}}$	L; L/T; T
$p, r_c;\ b_\sigma;\ y$	U; $\mathrm{L^{-1}}$; U
$a \times b;\ f, z;\ J$	$\mathrm{A^2}$; UL/T; UL/(TA)
$\lambda_g;\ \kappa_{\mathrm{in}};\ K$	$\mathrm{Q/A^2}$; UL/(TQ); $\mathrm{UL/(TA^2)}$
$\alpha_N;\ \beta_R;\ \eta_N, \eta_R$	$\mathrm{U^2L/A^2}$; $\mathrm{T^2/L}$; 1
$H_N, H_C, H_R;\ e;\ j, P_{NC}, P_{CR}$	$\mathrm{U^2L}$; $\mathrm{U^2}$; $\mathrm{U^2L/T}$

Table 2: Mathematical commensurability without SI calibration.

The local work identity has dimension $\mathrm{U^2/T}$, and its spatial integral is power. Restoring a native amplitude unit gives $[k_i] = \mathrm{A^2}$, $[\varepsilon] = \mathrm{A^{-2}}$ and dimensionless native g, λ; the fixed tuple below keeps its normalized meaning. Native step number has no time unit. Dimensional consistency does not physically calibrate the trial storages or cadence.

2 Reciprocal power without singular normalization

Proposition 2.1 (Homogeneous completion)**.** *With* $[v]_\times h = v \times h$*,*

$$J(x) = K\begin{bmatrix} -[b]_\times & [a]_\times \end{bmatrix}, \qquad J(x)x = 2f(x). \tag{2.1}$$

If $w = K^T y$ *and* $\mu = \eta_N/(2\alpha_N)$*, the core equation is equivalently*

$$J^T y = \begin{pmatrix} b \times w \\ w \times a \end{pmatrix}, \quad \dot a = \mu w \times b, \quad \dot b = -\mu w \times a, \quad \dot\Omega = -i\mu w \times \Omega. \tag{2.2}$$

Proof. Differentiation gives $D(a \times b)[\delta a, \delta b] = \delta a \times b + a \times \delta b$, proving the Jacobian and Euler identity. Transposing each skew cross-product matrix gives the two blocks in (2.2). Consequently $x^T J^T y = 2f^T y$; the factor one-half in (1.6) is necessary for the account below. □

Define the signed powers, density and flux

$$P_{NC} = \eta_N f(x)^T y, \qquad P_{CR} = \eta_R z^T y, \qquad e = \tfrac{1}{2}(p^2 + r_c^2), \quad j = cpr_c. \tag{2.3}$$

At strong solution regularity, multiplication of the two carrier equations by p, r_c gives the complete local identity

$$\partial_t e + \partial_s j = p\mathcal{B}\{\eta_N f(x) - \eta_R z\}. \tag{2.4}$$

Periodic integration removes $c\int \partial_s(pr_c)\,\mathrm{d}s$. Adjointness and (2.1) then give

$$\dot H_N = -P_{NC}, \qquad \dot H_C = P_{NC} - P_{CR}, \qquad \dot H_R = P_{CR}, \qquad \dot H_{\mathrm{tot}} = 0. \tag{2.5}$$

The receiver term is actual negative loading in the carrier equation, not a stock reconstructed from a residual. For every finite exchange window $[a_0, b_0]$, the integrated accounts are

$$\begin{aligned} \Delta H_N &= -\int_{a_0}^{b_0} P_{NC}\,\mathrm{d}t, \\ \Delta H_C &= \int_{a_0}^{b_0} (P_{NC} - P_{CR})\,\mathrm{d}t, \\ \Delta H_R &= \int_{a_0}^{b_0} P_{CR}\,\mathrm{d}t. \end{aligned} \tag{2.6}$$

Section 3 justifies these identities for the coupled mild solution as well as the strong solution. At periodic H^1 regularity, the local identity is distributional and point-cut fluxes have continuous spatial representatives. For arbitrary L^2 preparations we claim the global account and bounded port averages, not unrestricted point traces.

The feedback is polynomial in x and bounded linear in the carrier ports. There is no division by intensity, chirality or a state/output norm. The all-zero state is stationary. At $x = 0$, both f and J vanish, so even a preloaded carrier cannot charge native zero. At $p = r_c = z = 0$ with $f(x) \neq 0$, the carrier begins to change. At $z = 0$, $\dot H_R = 0$ at that instant although $\dot z$ can be nonzero and subsequent quadratic storage can grow. A vanishing port output at one instant need not remain zero.

The power identity does not uniquely select the completion: an additional sufficiently regular core term h satisfying $x \cdot h = 0$ would leave $\dot H_N$ unchanged while potentially altering other dynamics. No such term is added, and no classification is claimed.

3 The closed nonlinear solution problem

The exogenous C1 theorem alone does not prove well-posedness after closing the feedback. We give the coupled argument, using its inherited periodic translation group. [2, carrier theorem]

Theorem 3.1 (Global exchange evolution)**.** *For fixed finite allowed parameters and finite-width profiles, every preparation in*

$$\mathcal{H} = \mathbb{R}^6 \times L^2_{\mathrm{per}}(0, L_c)^2 \times \mathbb{R}^2 \tag{3.1}$$

has a unique mild solution of (1.6) in $C(\mathbb{R}; \mathcal{H})$. The core and receiver are C^1 in exchange time. The flow conserves H_{tot}, satisfies (2.6), and is locally Lipschitz in preparation on bounded energy sets over every finite time interval. For periodic H^1 carrier preparation,

$$(p, r_c) \in C(\mathbb{R}; H^1_{\mathrm{per}})^2 \cap C^1(\mathbb{R}; L^2)^2, \tag{3.2}$$

with continuous dependence also in this preparation topology. The local work identity (2.4) then holds distributionally. These statements concern exchange time, not a uniform bound over infinitely many native jumps.

Proof. Ports and free group. Integrating each affine half of a tent yields

$$\int b_\sigma \, \mathrm{d}s = 1, \quad \|b_\sigma\|_2^2 = m := \frac{2}{3\varepsilon_c}, \quad \|b_\sigma'\|_2^2 = \frac{2}{\varepsilon_c^3}, \quad M_B := \mathcal{B}^*\mathcal{B} = mI_2. \tag{3.3}$$

There are no delta terms in the first weak derivative because the tent is continuous. Disjoint supports remove the cross terms, also for weak derivatives up to null endpoints. Hence $\|\mathcal{B}\|_{\mathbb{R}^2 \to L^2} = \|\mathcal{B}^*\|_{L^2 \to \mathbb{R}^2} = \sqrt{m}$, and $\mathcal{B} : \mathbb{R}^2 \to H^1$ has squared norm $m_1 = m + 2/\varepsilon_c^3$ in the usual mathematical H^1 norm. This auxiliary regularity norm is not a different storage.

Put $v = (p, r_c)$ and $A_C v = (-c(r_c)_s, -cp_s)$ with domain $H^1_{\mathrm{per}}{}^2$. In the orthogonal coordinates $(p \pm r_c)/\sqrt{2}$, its group $S(t)$ translates by $\mp ct$. Periodic translation is an isometry in L^2 and H^1. Strong continuity follows first for trigonometric polynomials, then by density and the isometry bound. Its generator maps H^1 to L^2 by the displayed spatial derivative. Let $U(t) = \mathrm{diag}(\mathrm{Id}, S(t), \mathrm{Id})$. It is isometric in the norm

$$\|X\|_E^2 = \alpha_N |x|^2 + \|v\|_2^2 + \beta_R |z|^2 = 2H_{\mathrm{tot}}(X), \tag{3.4}$$

which is equivalent to the product norm at fixed positive scales.

Local nonlinear construction before conservation. Let $G(X)$ collect the core feedback, the forcing $(\mathcal{B}(\eta_N f - \eta_R z), 0)$ and receiver feedback. With $k_0 = \|K\|$, cross-product inequalities give

$$|f(x)| \le \tfrac{1}{2} k_0 |x|^2, \qquad \|J(x)\| \le k_0 |x|, \qquad |y| \le \sqrt{m}\, \|p\|_2 . \tag{3.5}$$

For example $|\delta a \times b + a \times \delta b| \le |x|\,|\delta x|$. The finite-dimensional polynomial factors and bounded ports therefore make G bounded and Lipschitz on every bounded energy ball. Seek the closed solution from

$$X(t) = U(t)X_0 + \int_0^t U(t-s) G(X(s)) \, \mathrm{d}s. \tag{3.6}$$

On the complete space $C([0, T]; \mathcal{H})$, take a ball of radius r about $U(t)X_0$. If M, L bound G and its Lipschitz constant on a state ball containing it, $TM \le r$ and $TL < 1$ make the integral map a self-map and contraction. Its fixed point is the local solution; the same construction works for negative times. The local time can be chosen from an energy bound on X_0 alone.

For two solutions staying in a common bounded ball, the isometry yields, for $t \ge 0$,

$$\left\| X(t) - \widetilde{X}(t) \right\|_E \le \left\| X_0 - \widetilde{X}_0 \right\|_E + L \int_0^t \left\| X(s) - \widetilde{X}(s) \right\|_E \mathrm{d}s \le e^{Lt} \left\| X_0 - \widetilde{X}_0 \right\|_E . \tag{3.7}$$

The scalar bound follows by iteration of the integral inequality. This proves uniqueness and local dependence independently of any formal energy cancellation.

Strong preparation and legitimate work. Along a local energy-space solution, $q(t) = \eta_N f(x(t)) - \eta_R z(t)$ is continuous. For $v_0 \in {H^1_{\mathrm{per}}}^2$, the carrier Duhamel formula thus belongs to $C_t H^1$, and for either time direction

$$\|v(t)\|_{H^1} \le \|v_0\|_{H^1} + \sqrt{m_1} \int_0^{|t|} |q(\pm s)| \, \mathrm{d}s. \tag{3.8}$$

The translation generator maps H^1 to L^2; differentiating the formula in L^2 gives (3.2) on the local interval. The finite-dimensional coordinates are C^1 because their port-dependent right sides are continuous. In one dimension, periodic H^1 functions have continuous representatives and a weak product rule. Accordingly the strong computation (2.4)–(2.6) is valid: the derivative of the squared L^2 norm exists and the integrated periodic derivative vanishes.

Energy-space work by approximation. Approximate only the carrier preparation by periodic H^1 functions converging in L^2. Their energy norms are bounded, so the local contraction interval is common. Estimate (3.7) gives convergence in $C_t\mathcal{H}$. The stocks converge by continuity of squared norms, while $f(x)$, z and $\mathcal{B}^* p$ converge uniformly on that interval. Thus all finite-time work integrals in (2.6) converge. Conservation and the integrated accounts pass to the mild solution and persist on successive local intervals. Here the work integrands are continuous, so the global integrated accounts even have continuous time derivatives; this does not require pointwise L^2 carrier differentiation.

Global restart and separate regularity bound. If $E_0 = H_{\mathrm{tot}}(X_0)$, the proved conservation gives

$$|x(t)| \le \sqrt{2E_0/\alpha_N}, \qquad \|v(t)\|_2 \le \sqrt{2E_0}, \qquad |z(t)| \le \sqrt{2E_0/\beta_R}. \tag{3.9}$$

The local restart time consequently has a uniform positive lower bound along the solution. No finite forward or backward maximal endpoint is possible. Estimate (3.7) also has finite-interval uniform constants for bounded initial energy sets.

For the stronger preparation, there is a distinct forcing estimate

$$|q(t)| \le \eta_N k_0 E_0/\alpha_N + \eta_R \sqrt{2E_0/\beta_R} =: Q_0, \qquad \|v(t)\|_{H^1} \le \|v_0\|_{H^1} + |t|\sqrt{m_1} Q_0. \tag{3.10}$$

An L^2 energy bound alone would not prove this H^1 estimate. For two preparations, the H^1 Duhamel difference is bounded by the initial H^1 difference plus $\sqrt{m_1} \int |q - \widetilde{q}|$. Local Lipschitzness of f and (3.7) bound that integral, proving finite-interval H^1 dependence. This completes all claims. □

The proof uses the complete-space contraction principle with its explicit local bounds, the translation group and the displayed scalar integral inequality. It does not infer existence from the name of a general interconnection framework. There is no uniform infinite-time H^1 estimate, no asserted H^2 regularity at tent corners and no uniform bound as $\varepsilon_c \downarrow 0$. Port averages remain L^2-bounded; point-cut claims retain the stronger trace hypothesis.

4 Nonzero supply, receiving and prepared credit

Proposition 4.1 (Empty-carrier launch)**.** *Let $\eta_N, \eta_R > 0$, $p_0 = r_{c,0} = z_0 = 0$ and $f_0 = f(x_0) \neq 0$. As $t \downarrow 0$,*

$$p(t) = \eta_N t \mathcal{B} f_0 + o(t) \quad \text{in } L^2, \qquad r_c(t) = o(t), \tag{4.1}$$

$$y(t) = \eta_N t M_B f_0 + o(t), \qquad x(t) - x_0 = O(t^2), \tag{4.2}$$

$$H_N(t) - H_N(0) = -\frac{\eta_N^2 t^2}{2} f_0^T M_B f_0 + o(t^2), \tag{4.3}$$

$$z(t) = \frac{\eta_N \eta_R t^2}{2\beta_R} M_B f_0 + o(t^2), \tag{4.4}$$

$$H_R(t) = \frac{\eta_N^2 \eta_R^2 t^4}{8\beta_R} \|M_B f_0\|^2 + o(t^4). \tag{4.5}$$

Such an x_0 exists inside the inherited coherent domain for every fixed yaw and positive inherited gain. Native supply and receiver gain are strict for all sufficiently small positive time.

Proof. For $\Omega_0 = (1, e^{i\phi}, 1)$ with $0 < \phi < \pi/12$, all moduli are one and the ordinary phase width is ϕ. This lies inside the old coherent domain. [4] Its chirality is $(-\sin\phi, 0, \sin\phi)$, giving

$$f_{0,\pm} = \frac{\kappa_{\text{in}} \lambda_g \sin\phi}{\sqrt{7}} (-2\sin\delta_g \pm \sqrt{3}). \tag{4.6}$$

The difference of the two entries is $2\sqrt{3}\kappa_{\text{in}}\lambda_g \sin\phi/\sqrt{7} \neq 0$, so both cannot vanish at any fixed yaw.

Empty carrier preparation belongs to H^1. Theorem 3.1 gives $p'(0) = \eta_N \mathcal{B} f_0$ and $r_c'(0) = 0$ in L^2, proving (4.1). Applying bounded $\mathcal{B}^*$ proves the port estimate. Since $x'(0) = 0$, the core equation with $y(t) = O(t)$ gives $x(t) - x_0 = O(t^2)$ and hence $f(x(t)) = f_0 + O(t^2)$. Integration of $-P_{NC}$ proves (4.3). Integrating the receiver equation gives (4.4); squaring it in H_R proves (4.5). Also $H_C(t) = \eta_N^2 t^2 f_0^T M_B f_0/2 + o(t^2)$. Since $M_B = mI_2$ with $m > 0$, the native debit and receiver gain have strictly positive coefficients in magnitude. More specifically P_{NC} has a positive leading term of order t, and

$$P_{CR}(t) = \frac{\eta_N^2 \eta_R^2 t^3}{2\beta_R} \|M_B f_0\|^2 + o(t^3) > 0 \tag{4.7}$$

for sufficiently small positive t. □

The argument uses L^2 first derivatives and integrals of continuous outputs; no smooth second derivative of a tent is presumed. Receiver loading is present in the exact system throughout. Because the two couplings are collocated, these estimates are neither arrival delays to a separate aperture nor a whole-circuit return. No trajectory is numerically evolved.

Proposition 4.2 (Prepared credit and reversible receiving)**.** *For $f_0 \neq 0$, $\eta_N > 0$ and $T_* > 0$ with time units, the preparations*

$$p_0 = \pm T_* \mathcal{B} f_0, \qquad r_{c,0} = z_0 = 0 \tag{4.8}$$

give

$$P_{NC}(0) = \pm \eta_N T_* f_0^T M_B f_0. \tag{4.9}$$

The plus preparation debits the native storage; the minus preparation credits it. For $\eta_R > 0$, admissible receiver preparations allow either sign of P_{CR}. The mathematical reversing involution is

$$\mathcal{R}(x, p, r_c, z) = (x, -p, r_c, z). \tag{4.10}$$

Proof. Equation (4.8) gives $y_0 = \pm T_* M_B f_0$, proving (4.9); continuity preserves each strict sign on a short interval. Each chosen $y_0 \in \mathbb{R}^2$ is realized by $p_0 = \mathcal{B} M_B^{-1} y_0 \in H^1$. Choosing $z_0 \cdot y_0 < 0$ discharges H_R, with equal positive carrier loading work; the opposite sign charges it. Substitution into (1.6) gives $F_{\mathrm{ex}}(\mathcal{R}X) = -\mathcal{R}F_{\mathrm{ex}}(X)$ for the full strong-state vector field. Uniqueness extends this reversal relation to mild solutions. □

The involution assigns no physical parity. The receiver has neither damping nor a one-way valve. Prepared negative power proves possible energetic return through this interface, not that a solution launched from an empty carrier later credits the native store. It also establishes no tagged cohort, spatial central route or completed circuit. Those questions are not solved by time reversal or a different preparation.

5 Dark states and a restrictive exchange invariant

If $f(x_0) = 0$ and $p_0 = r_{c,0} = z_0 = 0$, the constant state solves (1.6) and is unique. Aligned common-phase cores $\Omega_0 = e^{i\theta} q$ with real q have zero chirality and do not launch from this empty preparation. Projection-dark cores with $a \times b \in \ker K$ have the same conclusion. No noise, background source or alternative readout is added. A nonzero core with zero drive at one instant and a preloaded carrier can still have $J^T y \neq 0$; it is not automatically permanently dark. Native zero is different: $J(0) = 0$ makes it invariant under all allowed carrier/receiver preparations.

Theorem 5.1 (Invariant and net extraction capacity)**.** *The complex bilinear quantity*

$$Z_2(\Omega) = \sum_{i=1}^{3} \Omega_i^2 = |a|^2 - |b|^2 + 2i\, a \cdot b \tag{5.1}$$

is conserved throughout exchange. For every preparation of Theorem 3.1,

$$I^2 - |Z_2|^2 = 4|a \times b|^2, \tag{5.2}$$

$$H_N(t) \geq \frac{\alpha_N}{2} |Z_2(\Omega_0)|, \tag{5.3}$$

$$H_N(0) - H_N(t) \leq \frac{\alpha_N}{2} \{I_0 - |Z_2(\Omega_0)|\}. \tag{5.4}$$

If $I_0 > 0$ and the initial ordinary phase span is $\Delta_0 \leq \pi/12$, then

$$\frac{H_N(0) - H_N(t)}{H_N(0)} \leq 1 - \cos\Delta_0 \leq 1 - \frac{\sqrt{6} + \sqrt{2}}{4} \simeq 0.0340741737109317. \tag{5.5}$$

Proof. Equation (2.2) gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(|a|^2 - |b|^2) = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}t}(a \cdot b) = 0. \tag{5.6}$$

Indeed $a \cdot (w \times b) = -w \cdot (a \times b)$ and $b \cdot (w \times a) = w \cdot (a \times b)$ cancel in the difference. Each term in the derivative of $a \cdot b$ contains a vector dotted with its own cross product. Thus Z_2 is invariant. Expanding its squared modulus gives $I^2 - |Z_2|^2 = 4(|a|^2|b|^2 - (a \cdot b)^2) = 4|a \times b|^2 \geq 0$. Equivalently, $|\sum_i \Omega_i^2| \leq \sum_i |\Omega_i|^2$ by the triangle inequality. Conservation of Z_2 now proves (5.3)–(5.4).

For phases in an ordinary interval $[\theta_{\min}, \theta_{\max}]$ of width Δ_0, rotate Z_2 by $e^{-i(\theta_{\min} + \theta_{\max})}$. Each doubled phase then lies within Δ_0 of the new axis, so

$$\mathrm{Re}\{e^{-i(\theta_{\min} + \theta_{\max})} Z_2(\Omega_0)\} = \sum_i s_i \cos(2\theta_i - \theta_{\min} - \theta_{\max}) \geq I_0 \cos\Delta_0. \tag{5.7}$$

Combining this with (5.4) proves (5.5). □

The ceiling is approximately **3.4074% of initial core trial storage**, not a diversion share. It bounds net extraction $\int P_{NC}\,\mathrm{d}t$ during an exchange-only interval. The signed native change can instead be a gain. The result is not an attainability theorem and does not bound gross repeated positive work while storage is returned and resupplied. The fixed projection and dynamics may further restrict accessible transfer.

The initial phase-span restriction continues to imply the ceiling after the state leaves that sector, because Z_2 itself is conserved. At $I_0 = 0$ the normalized expression is undefined; the unnormalized statements and invariant zero state remain valid. At $\Delta_0 = 0$, $|Z_2| = I_0$, so positive net extraction is impossible; with empty carrier/receiver, total conservation forces complete inactivity. In Proposition 4.1, $|Z_2| = \sqrt{9 - 8\sin^2\phi} < 3$, consistent with positive short-time supply. Native F jumps can change Z_2 and reset this exchange-only bound. No lifetime hybrid capacity or conservation follows. Z_2 is complex bilinear, not the historical spatial/staged Z observation.

Proposition 5.2 (Common-phase equivariance)**.** *At fixed carrier and receiver state, exchange is equivariant under $\Omega \mapsto e^{i\theta}\Omega$; the drive and carrier/receiver evolution are unchanged by this common-phase transformation.*

Proof. The real-coordinate action Q_θ sends (a, b) to $(\cos\theta\, a - \sin\theta\, b, \sin\theta\, a + \cos\theta\, b)$. Its cross product is $a \times b$, hence $f(Q_\theta x) = f(x)$ and $J(Q_\theta x) = J(x)Q_\theta^T$. Orthogonality makes the core vector field transform by Q_θ while the remaining right sides are unchanged. Uniqueness gives the flow claim. Z_2 transforms by $e^{2i\theta}$, preserving its modulus. □

This candidate deliberately uses common-phase-invariant chirality. It need not use all phase information recovered by the earlier reconstruction result and supplies no common-phase-sensitive outlet selector. [3] Raw channel C, its ambient attachment, based face vectors and a spatial transfer law remain different objects.

6 The unchanged native map and ordered hybrid

Let $L_N = \mathbf{1}\mathbf{1}^T - 3\mathrm{Id}$ and retain the exact tuple

$$\varepsilon = \frac{1}{10}, \quad g = \frac{1}{60}, \quad \lambda = \frac{1}{300}, \quad k = \left(\frac{999}{1000}, 1, \frac{1001}{1000}\right). \tag{6.1}$$

With $s_i = |\Omega_i|^2$, the inherited simultaneous prestage and synchronization are [4, 5]

$$d_{\mathrm{pre},i} = \varepsilon(k_i - s_i)\Omega_i + g\sum_{j\neq i}(\Omega_j - \Omega_i), \qquad \widetilde{\Omega} = \Omega + d_{\mathrm{pre}}, \tag{6.2}$$

$$\delta_i = \lambda\sum_{j\neq i}\sin\{3[\mathrm{Arg}_0\,\widetilde{\Omega}_j - \mathrm{Arg}_0\,\widetilde{\Omega}_i]\}, \qquad F_i(\Omega) = e^{i\delta_i}\widetilde{\Omega}_i. \tag{6.3}$$

Here $\mathrm{Arg}_0(0) = 0$, and exact zero components remain zero after synchronization. Every prestage kick uses the same original state; every phase kick uses the same prestage. The source bypasses phase extraction at $\lambda = 0$, although the present tuple has nonzero λ. No normalization, coefficient change or continuous-gradient replacement is made. The prestage increment is not $F(\Omega) - \Omega$.

Synchronization preserves each prestage modulus. Expanding $|\Omega_i + d_{\mathrm{pre},i}|^2$ and summing gives the exact edge-once identity

$$I(F\Omega) - I(\Omega) = 2\varepsilon\sum_i s_i(k_i - s_i) - 2g\sum_{i<j}|\Omega_i - \Omega_j|^2 + \|d_{\mathrm{pre}}\|^2. \tag{6.4}$$

Each undirected mismatch appears at two endpoints. For $h_i = (k_i - s_i)\Omega_i$, the full finite-step square is

$$\|d_{\mathrm{pre}}\|^2 = \varepsilon^2 \|h\|^2 + 2\varepsilon g \,\mathrm{Re}(h^\dagger L_N \Omega) + g^2 \|L_N \Omega\|^2 , \tag{6.5}$$

$$\|L_N \Omega\|^2 = 3 \sum_{i<j} |\Omega_i - \Omega_j|^2. \tag{6.6}$$

The mixed term and total increment have no predetermined sign. These identities include zero-prestage strata and retain the exact finite-step remainder.

Proposition 6.1 (Exact hybrid account and recovery)**.** *Let Φ_h be the exact exchange flow of Theorem 3.1. For a new symbolic $h_{\mathrm{ex}} > 0$, define, in this order,*

$$X_{n+1} = \Phi_{h_{\mathrm{ex}}}(F_{\mathbb{R}}(x_n), p_n, r_{c,n}, z_n). \tag{6.7}$$

Then

$$H_{\mathrm{tot}}(X_{n+1}) - H_{\mathrm{tot}}(X_n) = \frac{\alpha_N}{2}[I(F\Omega_n) - I(\Omega_n)]. \tag{6.8}$$

At $\eta_N = \eta_R = 0$, the sampled native state follows exactly F, the carrier follows source-off C1 and the receiver is constant.

Proof. The native jump changes only x; its storage increment is the right side of (6.8). Exchange preserves the total thereafter. With both couplings zero, x, z are constant during exchange and the carrier is the free translation group, so sampling retains exactly the native update. After n steps the carrier has evolved freely for nh_{ex}. □

This is F first and exchange second, not a numerical integrator. The right side is the *native baseline storage increment*, not carrier/receiver work or a previously identified external source. Finite-step total changes sum these increments. No hidden reservoir compensates them, so the full hybrid is not automatically a closed energy-conserving physical system.

The inherited potential is

$$V = \varepsilon \sum_i \left(\frac{s_i^2}{4} - \frac{k_i s_i}{2} \right) + \frac{g}{2} \sum_{i<j} |\Omega_i - \Omega_j|^2, \qquad d_{\mathrm{pre}} = -\nabla_{\mathbb{R}^6} V. \tag{6.9}$$

Its accepted descent result applies to F at (6.1) on

$$\mathcal{D} = \{9/10 \le |\Omega_i| \le 11/10, \text{ ordinary phase width } \le \pi/12\}, \tag{6.10}$$

with arbitrary common orientation; it includes forward invariance and persistent nonzero prestages. [4] It is not a descent theorem for H_N or the coupled dynamics. For the aligned state $(1, 1, 1)$, synchronization is inactive and exactly

$$F(1,1,1) = (9999/10000, 1, 10001/10000), \qquad I(F(1,1,1)) - 3 = \frac{1}{50000000} > 0. \tag{6.11}$$

Thus H_N can increase where the distinct V theorem applies; real Z_2 also changes. This is a rational formula evaluation, not a trajectory experiment. Potential descent does not assign a recipient for native baseline change.

6.1 Limits, cadence and finite-index qualifications

If $\eta_R = 0$, z is constant and has no load, while native/carrier exchange can remain active. If $\eta_N = 0$, the native state is constant during exchange but still undergoes F at each jump; a nonzero receiver can interact with the carrier. Clamping x to an external history, suppressing its feedback and disabling the receiver gives exogenous C1 driven by $\eta_N f(x_{\text{history}})$. At $\eta_N = 1$ this is exactly the inherited chiral drive amplitude. An imposed hold reproduces that comparison protocol, but retaining drive while suppressing debit needs an external work account.

At active coupling the drive is continuously recomputed from x, whose feedback uses current carrier response. The older external sample/hold model is therefore not identical to (6.7). The cadence h_{ex} is not a derived physical period, a geometric shape parameter or the observer clock. The old source-off period L_c/c need not be a period of the coupled flow: absence of external forcing does not make $\mathcal{B}(\eta_N f - \eta_R z)$ zero. If the complete internal forcing vanishes throughout an interval, the source-off theorem applies on that hypothesis. [3]

Every finite ideal native state has a finite polynomial prestage and a defined finite synchronized image under the zero convention. The global exchange theorem then supplies the next interval. Induction establishes every finite hybrid iterate, including after departure from $\mathcal{D}$, without an infinite-index uniform bound.

Regular dependence is separate. Where every encountered prestage component is nonzero, F is smooth: sine of three phase differences can be expressed using normalized nonzero complex components, so an argument branch cut alone is not an obstruction. For finitely many such jumps, their moduli stay bounded away from zero in a sufficiently small preparation neighborhood; composition with (3.7) gives local continuous/Lipschitz dependence. At zero-prestage strata the convention defines F but does not supply general continuity or smoothness for neighbors using that phase. Exchange smoothness does not remove this native qualification.

Universal preservation of the old coherent domain is false for active native coupling. Appendix A gives two distinct accepted arguments: an exchange boundary witness and a sampled-hybrid witness with an admissible prepared carrier and short cadence. Neither says every preparation or cadence exits. Consequently the old convergence, quotient/closure and V-descent results remain on their original hypotheses rather than transferring automatically to this extension.

7 Transfer counts and receiver ratios

On a common finite window, define the matched per-port powers

$$P_{N,\sigma} = \eta_N f_\sigma(x) y_\sigma, \qquad P_{R,\sigma} = \eta_R z_\sigma y_\sigma. \tag{7.1}$$

Take positive/negative parts *before* summing the two ports:

$$\begin{aligned} Q_{N\to C} &= \int \sum_\sigma [P_{N,\sigma}]_+ \,\mathrm{d}t, \quad Q_{C\to N} = \int \sum_\sigma [-P_{N,\sigma}]_+ \,\mathrm{d}t, \\ Q_{C\to R} &= \int \sum_\sigma [P_{R,\sigma}]_+ \,\mathrm{d}t, \quad Q_{R\to C} = \int \sum_\sigma [-P_{R,\sigma}]_+ \,\mathrm{d}t. \end{aligned} \tag{7.2}$$

For simultaneous powers $A, -A$ with $A > 0$, this convention counts A each way; a positive part taken after summation would count zero. These finite work amounts are not geometric cut throughputs, tagged cohorts or per-pass toroidal efficiencies. The carrier cut flux j and these distributed interface powers have different definitions. [3]

For hybrid windows $[t_a, t_b]$, choose post-jump endpoint values if an endpoint is a jump, and count jumps $t_a < t_j \le t_b$. The pre-jump samples in (6.7) can instead use their matching left

limits and included jumps. Put

$$B_N = \sum_{t_a < t_j \le t_b} \frac{\alpha_N}{2}[I(F\Omega(t_j^-)) - I(\Omega(t_j^-))]. \tag{7.3}$$

The exact window accounts are

$$\begin{aligned} \Delta H_N &= B_N - Q_{N\to C} + Q_{C\to N}, \\ \Delta H_C &= Q_{N\to C} - Q_{C\to N} - Q_{C\to R} + Q_{R\to C}, \\ \Delta H_R &= Q_{C\to R} - Q_{R\to C}, \qquad \Delta H_{\mathrm{tot}} = B_N. \end{aligned} \tag{7.4}$$

They follow by integrating (2.5) between jumps and adding (6.8). Native jumps change neither carrier nor receiver and insert no impulsive port work. For exchange-only windows $B_N = 0$.

Proposition 7.1 (Restricted retained-storage ratio)**.** *If the carrier and receiver are both initially empty and $Q_{N\to C} > 0$, then*

$$0 \le \frac{\Delta H_R}{Q_{N\to C}} \le 1. \tag{7.5}$$

This holds for exchange-only windows and for finite hybrid windows with complete counting as above. If also $Q_{R\to C} = 0$, the same bound holds for $Q_{C\to R}/Q_{N\to C}$.

Proof. From (7.4),

$$\Delta H_R = Q_{N\to C} - Q_{C\to N} - \Delta H_C. \tag{7.6}$$

Empty initial stores give $\Delta H_R = H_R(t_b) \ge 0$ and $\Delta H_C = H_C(t_b) \ge 0$, while $Q_{C\to N} \ge 0$. This proves (7.5). Native jumps do not alter this carrier-plus-receiver account. If receiver backflow is zero, $Q_{C\to R} = \Delta H_R$. □

The supply denominator is positive for sufficiently short positive time in Proposition 4.1; it can be zero for dark or decoupled states, making the ratios undefined. The first ratio measures signed retained trial storage and the second gross receiving work. Gross receiving can recount back-and-forth transfer; without its extra no-backflow condition the retained-ratio proof is not a bound on gross receiving. Native baseline increments affect attribution to initial native stock and invalidate carrying the exchange-only capacity across jumps; they do not by themselves invalidate (7.5) under its actual start conditions.

There is no universal $[0, 1]$ bound for arbitrary preparation. An initial carrier stock can be released, and a receiver can have negative retained change. These possibilities have admissible small-window witnesses. Choose $f_0 \neq 0$, $\eta_N, \eta_R > 0$, $y_0 = T_* M_B f_0$ and $z_0 = A f_0$, with A dimensionless. The carrier preparation $p_0 = \mathcal{B} M_B^{-1} y_0$ realizes these ports. The initial supply rate is $\eta_N T_* f_0^T M_B f_0 > 0$, and the receiver rate is $\eta_R A T_* f_0^T M_B f_0$. For $A > \eta_N/\eta_R$, both small-window ratios tend to $\eta_R A/\eta_N > 1$; for $A < 0$, the retained ratio is negative. The signs follow analytically by continuity, without a parameter scan or evolved trajectory.

Prepared stock, backflow, repeated work and baseline increments must accompany interpretation. No physical efficiency or diversion percentage follows. Irreversible disposal would require additional justified state/loading or constitutive assumptions; none is introduced. C6 remains an unloaded observation and is not the candidate receiver or its storage. [2, exterior equation and limitations]

8 Interpretation and remaining questions

The new receiver has an analytical storage location inside the enlarged account and a specified loading region on the two C1 profiles. It has no recovered spatial/material location. Collocation is not a path to six central openings. The inherited distinctions remain [3]:

- The selected host-notch triangles N_1^+, N_1^- differ from the six central broken boundaries G_k^+, G_k^-. The latter are joined two-segment arcs, not filled apertures with newly assigned normals.
- The offset one-dimensional circle is distinct from the intended tight toroid and opposed cones. The original close enclosing sphere is a geometric reference, distinct from the extended-line sphere, not a receiving wall.
- Supports remain stationary. Based native vectors, raw chirality and its ambient attachment are not travel trajectories; common phase, static shape parameter, native index, carrier time and observer clock are different variables.

The known circle excludes exact O. That does not exclude every possible finite central starting/meeting region. No such receiving/control region or intended-route registration has been selected. A circle alone neither specifies nor rules out a surrounding three-dimensional field. No closing chord, material attachment, six-port allocation, electromagnetic law or gravity–magnetism correspondence is derived here.

The author's outward–around–inward picture and tentative "15–20%, possibly more" motivate a question, not data or a fitting target. The exact coherent-core extraction ceiling in (5.5) has a different denominator and is not a competing diversion estimate. The candidate's positive result is explicit reciprocal native state feedback and dynamic reversible receiving with matching work. Its invariant, dark states and domain failures sharply limit what follows from that construction.

8.1 Evidence and conclusion

This paper restates the accepted investigation with its complete bounded arguments. [1] The preserved symbolic helper records 28 exact identities from that investigation; it was not rerun for manuscript preparation, and a recorded reviewer replay would not be 28 additional distinct identities. Symbolic checks do not replace the nonlinear existence proof or justify physical interpretation. The current standalone publication order records GPT's analytical acceptance; the separate current review, optional bundle and older supporting review were unavailable in the named local search. That gap is disclosed in the evidence guide. The investigation's historical availability statements are retained unchanged. Codex's manuscript fidelity/build/page review is distinct from GPT's underlying analytical review; no final-PDF GPT review or external peer review is claimed.

One explicit reciprocal chiral exchange candidate is therefore mathematically well posed, supplies and receives trial storage with matched balances, and has a restrictive exchange invariant and domain limits. The exact native map is recovered at zero coupling, while the full hybrid exposes its native baseline increment. This is neither adoption into the unchanged accepted kernel nor a derived physical energy law or demonstrated toroidal circulation. Delayed total native credit on the same solution after an empty launch remains open; prepared credit and reversal do not settle it. Irreversible receiving, accessible-capacity attainability and physical route registration are also separate questions. No successor investigation is carried out here, and no diversion percentage is fitted or established.

A Two distinct coherent-domain counterexamples

We retain both accepted witnesses because a substep exit alone would not prove that a sampled composition exits. Throughout, $\eta_N > 0$ and $\mu = \eta_N/(2\alpha_N)$.

Proposition A.1 (Exchange boundary exit)**.** *The domain $\mathcal{D}$ in (6.10) is not invariant under exchange for all admitted carrier preparations.*

Proof. Set $\Omega_0 = (1, e^{i\phi}, 1)$ at $\phi = \pi/12$, and choose $w_0 = K^T y_0 = -We_z$ with $W > 0$. This direction is available at every yaw: writing $\gamma = \kappa_{\text{in}}\lambda_g$, choose

$$y_0 = \frac{\sqrt{7}W}{2\sqrt{3}\gamma}(-1, 1), \quad p_0 = \mathcal{B}M_B^{-1}y_0, \quad r_{c,0} = z_0 = 0. \tag{A.1}$$

The carrier is H^1 and the native moduli are strictly inside their radial interval. Equation (2.2) and $\dot{\theta}_i = (a_i\dot{b}_i - b_i\dot{a}_i)/|\Omega_i|^2$ give

$$\dot{\theta}_1(0) = -\mu W\cos\phi, \qquad \dot{\theta}_2(0) = \mu W\cos\phi, \qquad \dot{\theta}_3(0) = 0. \tag{A.2}$$

For small positive time, θ_1 is the minimum and θ_2 the maximum. The right derivative of phase width is $2\mu W\cos\phi > 0$. Thus the initial exit is through the phase boundary, without a clamp or normalization. □

Proposition A.2 (Sampled hybrid witness)**.** *Over the allowed carrier preparations and symbolic positive cadences, the sampled hybrid* (6.7) *does not universally preserve* $\mathcal{D}$*.*

Proof. Start the hybrid step at $(1, 1, 1)$. Its native image is the positive real vector $q = (q_1, q_2, q_3)$ in (6.11). Take an H^1 profile $\bar{p}$ with $K^T\mathcal{B}^*\bar{p} = -We_z$, realized as in (A.1). Prepare $p_0 = R\bar{p}$, $r_{c,0} = z_0 = 0$, with dimensionless $R > 0$, and use fast time $\tau = Rt$. All model constants remain fixed.

On bounded τ intervals, the energy bound makes $\|p_R/R\|_2$ uniformly bounded as $R \to \infty$. The core equation and its integral inequality then give $|x_R(\tau/R)| \le |q|e^{C\tau}$ uniformly, and integration of the receiver equation gives $|z_R(\tau/R)| \le C\tau$. The carrier Duhamel formula divided by R has free term $S(\tau/R)(\bar{p}, 0) \to (\bar{p}, 0)$ uniformly on compact τ intervals by strong continuity. Its forcing term is $O(R^{-2})$ there: the forcing coefficient is bounded by the preceding core/receiver estimates, the interval has length $O(R^{-1})$, and we divide by R. Therefore

$$p_R(\tau/R)/R \longrightarrow \bar{p}, \qquad K^T y_R(\tau/R)/R \longrightarrow -We_z. \tag{A.3}$$

The finite-dimensional integral equation and a Lipschitz/Gronwall estimate give uniform core convergence to

$$\begin{aligned} &\frac{\mathrm{d}\Omega}{\mathrm{d}\tau} = i\mu W e_z \times \Omega, \\ &\Omega_1 = q_1\cosh(\mu W\tau) - iq_2\sinh(\mu W\tau), \\ &\Omega_2 = q_2\cosh(\mu W\tau) + iq_1\sinh(\mu W\tau), \qquad \Omega_3 = q_3. \end{aligned} \tag{A.4}$$

Choose the finite symbolic τ_* with $\tanh(\mu W\tau_*) = 1/4$. Its limiting phase width is

$$\arctan\frac{q_2}{4q_1} + \arctan\frac{q_1}{4q_2} > \pi/12. \tag{A.5}$$

To see the strict inequality without sampling, both q_2/q_1 and q_1/q_2 exceed $9/10$. The left side exceeds $2\arctan(9/40) > \pi/12$, since $\tan(\pi/24) = \sin(\pi/12)/(1 + \cos(\pi/12)) < 2/9 < 9/40$. The limiting components are nonzero, so uniform convergence preserves this strict phase excess for sufficiently large finite R. Selecting the allowed cadence $h_{\text{ex}} = \tau_*/R$ then puts the sampled hybrid output outside $\mathcal{D}$. □

This is an existential analytic preparation/cadence witness, not a numerical fast-time experiment or a claim that every fixed cadence, preparation or empty launch exits. At $\eta_N = 0$, native sampling still follows F and retains the old domain property. Neither counterexample modifies the older theorem on its own hypotheses.

References

[1] Tri-Octagon research programme. *Native reciprocal exchange: independent proof review.* NATIVE_RECIPROCAL_EXCHANGE_01, 3 November 2026. Retained original proof and accepted RX01–RX41 ledger. The publication authorization records GPT's analytical acceptance and all operative qualifications. The separate current review file was unavailable locally; its text is not reconstructed.

[2] H. F. Halldórsson. *Paired Driving and Exterior Identifiability on a Conditional Periodic Carrier.* Paper II, repository research edition v0.1; manuscript dated 19 October 2026. Immutable reading edition. Foundations, C1 theorem and work account, native hold protocol, unloaded C6 and limitations. The retained manuscript bears its historical draft label.

[3] H. F. Halldórsson. *Recirculation and Diversion Accounting on a Conditional Tri-Octagon Carrier.* Repository research edition v0.1, 3 November 2026. Immutable edition. Recurrence, distributed work versus cut transport, receiving-account gap and spatial interpretation.

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[5] Tri-Octagon research programme. *Native definitions as mathematical text*: `dynamics.py`, `readouts.py`, `z_diagnostics.py`, `face_state.py`. Immutable reading directory. Exact used file identities and pertinent functions are recorded in the edition's source inventory. No module was imported or run for this publication.

[6] A. van der Schaft. *Port-Hamiltonian nonlinear systems.* arXiv:2412.19673v1, 2024. Version-pinned primary text, Sections 1.1–1.2, equations (5)–(6) and the power-preserving interconnection discussion. Methodological context only; no full-text cache is redistributed.